FORECASTING

1

WHAT IS FORECASTING?

- Process of predicting a future event
- Underlying basis of all business decisions
  - Production
  - Inventory
  - Personnel
  - Facilities

2

FORECASTING TIME HORIZONS

- Short-range forecast
  - Up to 1 year, generally less than 3 months
  - Purchasing, job scheduling, workforce levels, job assignments, production levels
- Medium-range forecast
  - 3 months to 3 years
  - Sales and production planning, budgeting
- Long-range forecast
  - 3^{+} years
  - New product planning, facility location, research and development

3

DISTINGUISHING DIFFERENCES

- Medium/long range forecasts deal with more comprehensive issues and support management decisions regarding planning and products, plants and processes
- Short-term forecasts tend to be more accurate than longer-term forecasts

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TYPES OF FORECASTS

- Economic forecasts
  - Address business cycle – inflation rate, money supply, housing starts, etc.
- Technological forecasts
  - Predict rate of technological progress
  - Impacts development of new products
- Demand forecasts
  - Predict sales of existing products and services

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APPROACH TO DEMAND FORECASTING

- Understand the objective of forecasting
- Integrate demand planning and forecasting throughout the supply chain
- Understand and identify customer segments
- Identify major factors that influence the demand forecast
- Determine the appropriate forecasting technique
- Establish performance and error measures for the forecast.

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SEVEN STEPS IN FORECASTING

- Determine the use of the forecast
- Select the items to be forecasted
- Determine the time horizon of the forecast
- Select the forecasting model(s)
- Gather the data
- Make the forecast
- Validate and implement results

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THE REALITIES!

☑Forecasts are seldom perfect

☑Most techniques assume an underlying stability in the system

☑Product family and aggregated forecasts are more accurate than individual product forecasts

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FORECASTING APPROACHES

Qualitative Methods (Judge-mental)

☑Used when situation is vague and little data exist
- ☑New products
- ☑New technology

☑Involves intuition, experience
- ☑e.g., forecasting sales on Internet

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FORECASTING APPROACHES

Quantitative Methods (Statistical)

☑Used when situation is 'stable' and historical data exist
- ☑Existing products
- ☑Current technology

☑Involves mathematical techniques
- ☑e.g., forecasting sales of color televisions

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JURY OF EXECUTIVE OPINION

☑Involves small group of high-level experts and managers

☑Group estimates demand by working together

☑Combines managerial experience with statistical models

☑Relatively quick

☑'Group-think' disadvantage

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SALES FORCE COMPOSITE

☑Each salesperson projects his or her sales

☑Combined at district and national levels

☑Sales reps know customers' wants

☑Tends to be overly optimistic

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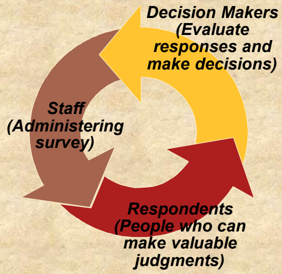
DELPHI METHOD

- Iterative group process, continues until consensus is reached
- 3 types of participants
  - Decision makers
  - Staff
  - Respondents



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CONSUMER MARKET SURVEY

- Ask customers about purchasing plans
- What consumers say, and what they actually do are often different
- Sometimes difficult to answer

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OVERVIEW OF QUANTITATIVE APPROACHES

1. Naive approach
2. Moving averages
3. Exponential smoothing
4. Trend projection
5. Linear regression

Time-Series Models (1–4)

Associative Model (5)

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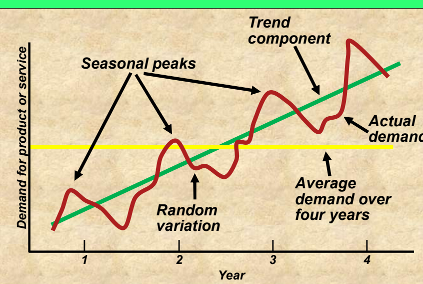
COMPONENTS OF DEMAND



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NAIVE APPROACH

- Assumes demand in next period is the same as demand in most recent period
  - e.g., If January sales were 68, then February sales will be 68
- Sometimes cost effective and efficient
- Can be good starting point

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MOVING AVERAGE METHOD

- MA is a series of arithmetic means
- Used if little or no trend
- Used often for smoothing
  - Provides overall impression of data over time

$$\text{Moving average} = \frac{\sum \text{demand in previous } n \text{ periods}}{n}$$

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MOVING AVERAGE EXAMPLE

Month	Actual Shed Sales	3-Month Moving Average
January	10	
February	12	
March	13	
April	16	(10 + 12 + 13)/3 = 11 $^2/_3$
May	19	(12 + 13 + 16)/3 = 13 $^2/_3$
June	23	(13 + 16 + 19)/3 = 16
July	26	(16 + 19 + 23)/3 = 19 $^1/_3$

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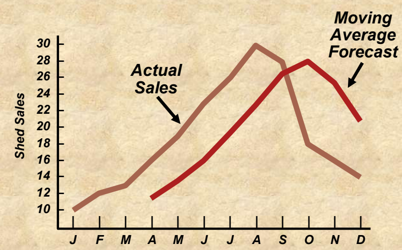
GRAPH OF MOVING AVERAGE



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WEIGHTED MOVING AVERAGE

- Used when trend is present
  - Older data usually less important
- Weights based on experience and intuition

$$\text{Weighted moving average} = \frac{\sum (\text{weight for period } n) \times (\text{demand in period } n)}{\sum \text{weights}}$$

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WEIGHTED MOVING AVERAGE

Weights Applied	Period
3	Last month
2	Two months ago
1	Three months ago
6	Sum of weights

Month	Actual Shed Sales	3-Month Weighted Moving Average
January	10	
February	12	
March	13	
April	16	[(3 x 13) + (2 x 12) + (10)]/6 = 12$^1/_6$
May	19	[(3 x 16) + (2 x 13) + (12)]/6 = 14$^1/_3$
June	23	[(3 x 19) + (2 x 16) + (13)]/6 = 17
July	26	[(3 x 23) + (2 x 19) + (16)]/6 = 20$^1/_2$

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POTENTIAL PROBLEMS WITH MOVING AVERAGE

- Increasing n smooth's the forecast but makes it less sensitive to changes
- Do not forecast trends well
- Require extensive historical data

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MOVING AVERAGE AND WEIGHTED MOVING AVERAGE

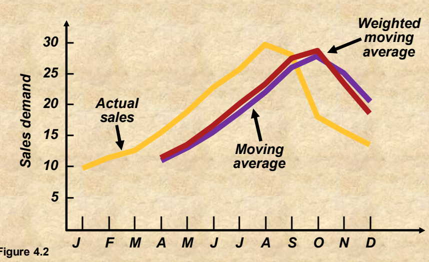

Figure 4.2

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- *The monthly demand for a company for the past 10 months is as per below. The manager has accumulated the data for ten months and wants to compute the three and five months moving averages.*

Month	Units
January	120
February	90
March	100
April	75
May	110
June	50
July	75
August	130
September	110
October	90

- *The manager wants to identify if there is a variation in the forecast in case he uses three months weighted moving average with the weights for latest three months before the month's demand to be forecasted as 0.17; 0.33 and 0.50 in sequence of n-3; n-2 and n-1 month. Draw a graph to show the variation.*

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EXPONENTIAL SMOOTHING

- ☑ Form of weighted moving average
  - ☑ Weights decline exponentially
  - ☑ Most recent data weighted most
- ☑ Requires smoothing constant (α)
  - ☑ Ranges from 0 to 1
  - ☑ Subjectively chosen
- ☑ Involves little record keeping of past data

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EXPONENTIAL SMOOTHING

New forecast = Last period's forecast
+ α (Last period's actual demand
– Last period's forecast)

$$F_t = F_{t-1} + \alpha(D_{t-1} - F_{t-1})$$

where F_t = *new forecast*
F_{t-1} = *previous forecast*
α = *smoothing (or weighting) constant* ($0 \le \alpha \le 1$)

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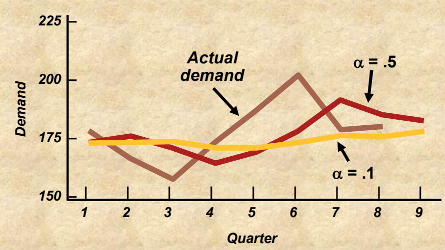
IMPACT OF DIFFERENT α



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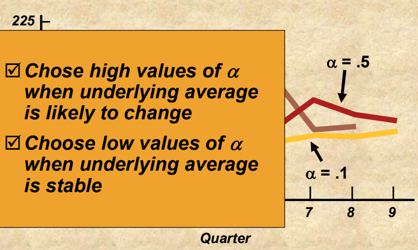
IMPACT OF DIFFERENT α



☑ Chose high values of α when underlying average is likely to change
☑ Choose low values of α when underlying average is stable

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CHOOSING α

The objective is to obtain the most accurate forecast no matter the technique

We generally do this by selecting the model that gives us the lowest forecast error

Forecast error = Actual demand - Forecast value

$= A_t - F_t$

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PROBLEM 1

- The monthly demand for units manufactured by the Acme Rocket Company has been as follows:

Month	Units
May	100
June	80
July	110
August	115
September	105
October	110
November	125
December	120

- Use the exponential smoothing method to forecast the number of units for June-January. The initial forecast for May was 105 units and $\alpha = 0.2$.

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Current Month	$F_{t+1} = \alpha D_t + (1 - \alpha) F_t$.	Forecast Month
May	0.2 x 100 + 0.8(105) = 104	June
June	0.2 x 80 + 0.8(104) = 99	July
July	0.2 x 110 + 0.8(99) = 101	August
August	0.2 x 115 + 0.8(101) = 104	September
September	0.2 x 105 + 0.8(104) = 104	October
October	0.2 x 110 + 0.8(104) = 105	November
November	0.2 x 125 + 0.8(105) = 109	December
December	0.2 x 120 + 0.8(109) = 111	January

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- The monthly demand for computers for MS computers for the past 12 months. The manager wants to use the exponential smoothing forecasts using smoothing constants equal to 0.30 and 0.50

Month	Units
January	37
February	40
March	41
April	37
May	45
June	50
July	43
August	47
September	56
October	52
November	55
December	54

- Use the exponential smoothing method to forecast the number of units for January next year. The initial forecast for January this year may be taken as equal to demand.

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COMMON MEASURES OF ERROR

***Mean Absolute Deviation* (*MAD*):**

$$MAD = \frac{\sum |Actual - Forecast|}{n}$$

***Mean Squared Error* (*MSE*):**

measures the vaiance of forcast error with standard normal distribution

$$MSE = \frac{\sum (Forecast\ Errors)^2}{n}$$

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COMMON MEASURES OF ERROR

***Mean Absolute Percent Error* (*MAPE*)**

$$MAPE = \frac{\sum_{i=1}^{n} 100|Actual_i - Forecast_i|/Actual_i}{n}$$

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COMPARISON OF FORECAST ERROR

Quarter	Actual Tonnage Unloaded	Rounded Forecast with α = .10	Absolute Deviation for α = .10	Rounded Forecast with α = .50	Absolute Deviation for α = .50
1	180	175	5.00	175	5.00
2	168				
3	159				
4	175				
5	190				
6	205				
7	180				
8	182				

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COMPARISON OF FORECAST ERROR

Quarter	Actual Tonnage Unloaded	Rounded Forecast with α = .10	Absolute Deviation for α = .10	Rounded Forecast with α = .50	Absolute Deviation for α = .50
1	180	175	5.00	175	5.00
2	168	175.5	7.50	177.50	9.50
3	159	174.75	15.75	172.75	13.75
4	175	173.18	1.82	165.88	9.12
5	190	173.36	16.64	170.44	19.56
6	205	175.02	29.98	180.22	24.78
7	180	178.02	1.98	192.61	12.61
8	182	178.22	3.78	186.30	4.30
			82.45		98.62

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COMPARISON OF FORECAST ERROR

$$MAD = \frac{\sum |deviations|}{n}$$

For α = .10

= 82.45/8 = 10.31

For α = .50

= 98.62/8 = 12.33

Quarter	Actual Tonnage Unloaded	Rounded Forecast with α = .10	Absolute Deviation for α = .10	Rounded Forecast with α = .50	Absolute Deviation for α = .50
				175	5.00
				177.50	9.50
				172.75	13.75
				165.88	9.12
				170.44	19.56
				180.22	24.78
				192.61	12.61
8	182	178.22	3.78	186.30	4.30
			82.45		98.62

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COMPARISON OF FORECAST ERROR

$$MSE = \frac{\sum (forecast\ errors)^2}{n}$$

For α = .10

= 1,526.54/8 = 190.82

For α = .50

= 1,561.91/8 = 195.24

Quarter	Actual Tonnage Unloaded	Rounded Forecast with α = .10	Absolute Deviation for α = .10	Rounded Forecast with α = .50	Absolute Deviation for α = .50
				175	5.00
				177.50	9.50
				172.75	13.75
				165.88	9.12
				170.44	19.56
				180.22	24.78
				192.61	12.61
8	182	178.22	3.78	186.30	4.30
			82.45		98.62
		MAD	10.31		12.33

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COMPARISON OF FORECAST ERROR

$$MAPE = \frac{\sum_{i=1}^{n} 100|deviation_i|/actual_i}{n}$$

For α = .10

= 44.75/8 = 5.59%

For α = .50

= 54.05/8 = 6.76%

Quarter	Actual Tonnage Unloaded	Rounded Forecast with α = .10	Absolute Deviation for α = .10	Rounded Forecast with α = .50	Absolute Deviation for α = .50
					5.00
					9.50
					13.75
					9.12
					19.56
					24.78
					12.61
8	182	178.22	3.78	186.30	4.30
			82.45		98.62
		MAD	10.31		12.33
		MSE	190.82		195.24

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COMPARISON OF FORECAST ERROR

Quarter	Actual Tonnage Unloaded	Rounded Forecast with α = .10	Absolute Deviation for α = .10	Rounded Forecast with α = .50	Absolute Deviation for α = .50
1	180	175	5.00	175	5.00
2	168	175.5	7.50	177.50	9.50
3	159	174.75	15.75	172.75	13.75
4	175	173.18	1.82	165.88	9.12
5	190	173.36	16.64	170.44	19.56
6	205	175.02	29.98	180.22	24.78
7	180	178.02	1.98	192.61	12.61
8	182	178.22	3.78	186.30	4.30
			82.45		98.62
		MAD	10.31		12.33
		MSE	190.82		195.24
		MAPE	5.59%		6.76%

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TREND PROJECTIONS

Fitting a trend line to historical data points to project into the medium to long-range

Linear trends can be found using the least squares technique

$$\hat{y} = a + bx$$

where $\hat{y}$ = computed value of the variable to be predicted (dependent variable)
a = y-axis intercept
b = slope of the regression line
x = the independent variable

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LEAST SQUARES METHOD

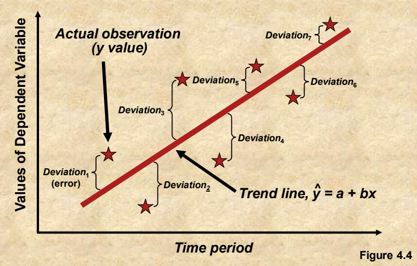

Figure 4.4

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LEAST SQUARES METHOD

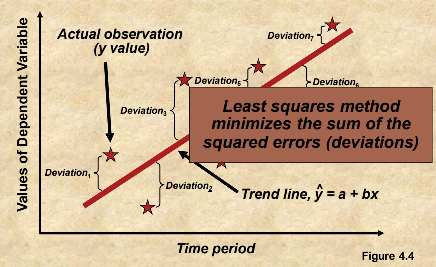

Figure 4.4

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LEAST SQUARES METHOD

Equations to calculate the regression variables

$$\hat{y} = a + bx$$

$$b = \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma x^2 - n\bar{x}^2}$$

$$a = \bar{y} - b\bar{x}$$

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LEAST SQUARES EXAMPLE

Year	Time Period (x)	Electrical Power Demand	x^2	xy
2001	1	74	1	74
2002	2	79	4	158
2003	3	80	9	240
2004	4	90	16	360
2005	5	105	25	525
2005	6	142	36	852
2007	7	122	49	854
	$\sum x = 28$	$\sum y = 692$	$\sum x^2 = 140$	$\sum xy = 3{,}063$
	$\bar{x} = 4$	$\bar{y} = 98.86$		

$$b = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2} = \frac{3{,}063 - (7)(4)(98.86)}{140 - (7)(4^2)} = 10.54$$

$$a = \bar{y} - b\bar{x} = 98.86 - 10.54(4) = 56.70$$

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LEAST SQUARES EXAMPLE

Year	Time Period (x)	Electrical Power Demand	x^2	xy
1999	1	74	1	74
			4	158
			9	240
			16	360
			25	525
			36	852
2005		122	49	854
	$\Sigma x = 28$	$\Sigma y = 692$	$\Sigma x^2 = 140$	$\Sigma xy = 3{,}063$
	$\bar{x} = 4$	$\bar{y} = 98.86$		

The trend line is

$$\hat{y} = 56.70 + 10.54x$$

$$b = \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma x^2 - n\bar{x}^2} = \frac{3{,}063 - (7)(4)(98.86)}{140 - (7)(4^2)} = 10.54$$

$$a = \bar{y} - b\bar{x} = 98.86 - 10.54(4) = 56.70$$

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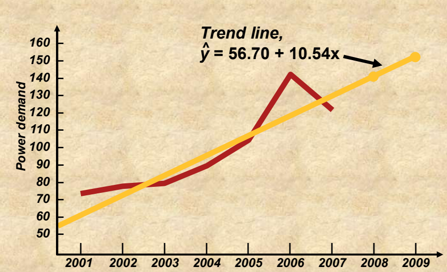
LEAST SQUARES EXAMPLE



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LEAST SQUARES REQUIREMENTS

1. We always plot the data to insure a linear relationship
2. We do not predict time periods far beyond the database
3. Deviations around the least squares line are assumed to be random

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ASSOCIATIVE FORECASTING

Used when changes in one or more independent variables can be used to predict the changes in the dependent variable

Most common technique is linear regression analysis

We apply this technique just as we did in the time series example

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ASSOCIATIVE FORECASTING

Forecasting an outcome based on predictor variables using the least squares technique

$$\hat{y} = a + bx$$

where $\hat{y}$ = *computed value of the variable to be predicted (dependent variable)*
a = *y-axis intercept*
b = *slope of the regression line*
x = *the independent variable though to predict the value of the dependent variable*

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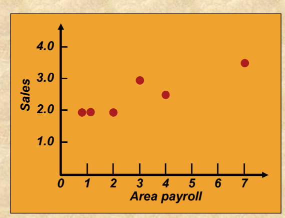
ASSOCIATIVE FORECASTING EXAMPLE

Sales ($ millions), y	Local Payroll ($ billions), x
2.0	1
3.0	3
2.5	4
2.0	2
2.0	1
3.5	7



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ASSOCIATIVE FORECASTING EXAMPLE

Sales, y	Payroll, x	x^2	xy
2.0	1	1	2.0
3.0	3	9	9.0
2.5	4	16	10.0
2.0	2	4	4.0
2.0	1	1	2.0
3.5	7	49	24.5
$\sum y = 15.0$	$\sum x = 18$	$\sum x^2 = 80$	$\sum xy = 51.5$

$\bar{x} = \sum x/6 = 18/6 = 3$ $\quad b = \dfrac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2} = \dfrac{51.5 - (6)(3)(2.5)}{80 - (6)(3^2)} = .25$

$\bar{y} = \sum y/6 = 15/6 = 2.5$ $\quad a = \bar{y} - b\bar{x} = 2.5 - (.25)(3) = 1.75$

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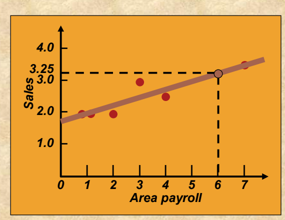
ASSOCIATIVE FORECASTING EXAMPLE

$\hat{y} = 1.75 + .25x$ $\qquad$ *Sales* = 1.75 + .25(*payroll*)

If payroll next year is estimated to be $6 billion, then:

Sales = 1.75 + .25(6)
Sales = $3,250,000



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FORECASTING IN THE SERVICE SECTOR

- Presents unusual challenges
  - Special need for short term records
  - Needs differ greatly as function of industry and product
  - Holidays and other calendar events
  - Unusual events

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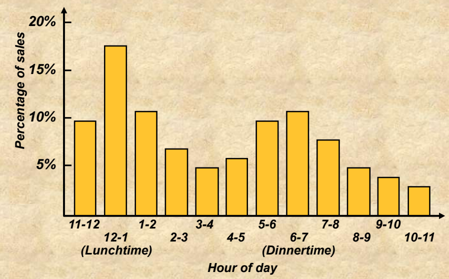
FAST FOOD RESTAURANT FORECAST



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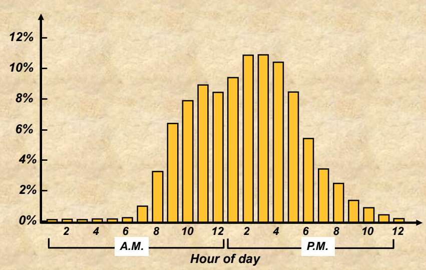
FEDEX CALL CENTER FORECAST



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Exponential smoothing is used to forecast automobile battery sales. Two value of α are examined, α=0.8 and α=0.5. Evaluate the accuracy of each smoothing constant. Which is preferable? (Assume the forecast for January was 22 batteries.) Actual sales are given below:

Month	Actual Battery Sales	Forecast
January	20	22
February	21	
March	15	
April	14	
May	13	
June	16	

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PROBLEM 1

- *The monthly demand for units delivered by a company is as per the table below:*
- *Use the exponential smoothing method to forecast the number of units for February-November. The initial forecast for January was 90 units. Do a comparison for α = 0.2 and α = 0.8 .*

Month	Units	Forecast 0.2	Forecast 0.8
January	100		
February	95		
March	105		
April	110		
May	100		
June	130		
July	90		
August	110		
September	100		
October	140		
November			

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PROBLEM 2

- *The Yearly demand for three years (quarterly) is given in the table below. Extract the trend component of the given data and predict the future demand for next year. monthly demand for units delivered by a company is as per the table below:*

Year	Quarter	Actual Demand (Y)	X	X*Y	X*X
1	1	360	1	360	1
1	2	438	2	876	4
1	3	359	3	1078	9
1	4	406	4		
2	1	393	5		
2	2	465	6		
2	3	387	7		
2	4	464	8		
3	1	505	9		
3	2	618	10		
3	3	443	11		
3	4	540	12		

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Year	Sales (Units)
2001	100
2002	110
2003	122
2004	130
2005	139
2006	152
2007	164

USE THE SALES DATA GIVEN BELOW TO DETERMINE:
(A) THE LEAST SQUARES TREND LINE
(B) THE PREDICTED VALUE FOR 2008 SALES.

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A manufacturer of critical components has the data of 18 months of demand and forecast as below: Compute the measures of forecast accuracy and comment on the usefulness of the forecasting system

Period	Demand	Forecast	Period	Demand	Forecast
1	120	109	10	109	104
2	114	118	11	123	110
3	130	132	12	119	119
4	124	110	13	130	124
5	97	110	14	125	110
6	95	105	15	119	90
7	100	98	16	120	95
8	110	95	17	90	75
9	109	104	18	95	65

PROBLEM 4

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